## $N = 785_{10}$

### Binary

| ÷2 | Q | R |
|---|---|---|
| 785 | 392 | 1 |
| 392 | 196 | 0 |
| 196 | 98 | 0 |
| 98 | 49 | 0 |
| 49 | 24 | 1 |
| 24 | 12 | 0 |
| 12 | 6 | 0 |
| 6 | 3 | 0 |
| 3 | 1 | 1 |
| 1 | 0 | 1 |

$785_{10} = \mathbf{1100010001_2}$

### Octal and Hexadecimal

| ÷8 | Q | R |
|---|---|---|
| 785 | 98 | 1 |
| 98 | 12 | 2 |
| 12 | 1 | 4 |
| 1 | 0 | 1 |

| ÷16 | Q | R |
|---|---|---|
| 785 | 49 | 1 |
| 49 | 3 | 1 |
| 3 | 0 | 3 |

$785_{10} = \mathbf{1421_8}$ $785_{10} = \mathbf{311_{16}}$

### Verification

| $2^9$ | $2^8$ | $2^7$ | $2^6$ | $2^5$ | $2^4$ | $2^3$ | $2^2$ | $2^1$ | $2^0$ |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 |

$512 + 256 + 16 + 1 = 785_{10}$ ✓

| $8^3$ | $8^2$ | $8^1$ | $8^0$ |
|---|---|---|---|
| 1 | 4 | 2 | 1 |

| $16^2$ | $16^1$ | $16^0$ |
|---|---|---|
| 3 | 1 | 1 |

$512 + 256 + 16 + 1 = 785_{10}$ ✓ $768 + 16 + 1 = 785_{10}$ ✓

### Grouping

001 | 100 | 010 | 001 = $\mathbf{1421_8}$ ✓
0011 | 0001 | 0001 = $\mathbf{311_{16}}$ ✓

### Error analysis

46 ÷ 2 gives remainders 0, 1, 1, 1, 0, 1 (top to bottom). The classmate read them top to bottom instead of bottom to top, so the answer came out reversed. $011101_2$ = 29, not 46.
Correct: $46_{10} = \mathbf{101110_2}$ (32 + 8 + 4 + 2 = 46 ✓)